

Smart Pantry Manager

A Java Android Application that Suggests Recipes Based Strictly on Leftover
Ingredients to Cut Food Waste

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1. Introduction

Every week households around the world buy food with good intentions, store it, forget about it, and eventually throw it away. The United Nations Environment Programme estimates that roughly one fifth of the food available to consumers is lost or wasted globally (UNEP 2021). In practice the waste rarely happens because people want to waste food; it happens because the half-onion in the fridge, the last two eggs and the tail-end of a bag of flour are invisible at the moment a person asks the everyday question: *"what can I cook tonight?"*. The result is a takeaway order or a fresh shop while perfectly good leftovers quietly expire.

Smart Pantry Manager is a native Android application, written in Java, that answers that question using only what the user already owns. The user records the ingredients they have at home — their *pantry* — with a quantity, a unit of measurement and an optional expiry date. The app then compares that pantry, ingredient by ingredient and quantity by quantity, against a built-in collection of twenty recipes, and suggests only the dishes the user can genuinely cook right now with no shopping trip required.

The defining feature of the app, and the piece of business logic treated as its core, is the **strict-matching rule**: a recipe appears in the suggestions list if and only if every single ingredient it requires is present in the pantry in at least the required quantity. A recipe needing five ingredients of which the pantry holds four is deliberately excluded — no "almost there" recipe is allowed to masquerade as a suggestion in the main list. The app does, however, offer a clearly separated bonus section that names recipes missing exactly one ingredient, because that small nudge ("you could make pancakes if you had milk") is precisely the kind of information that prevents the next bag of flour from being wasted.

The application is aimed at three overlapping groups of users: home cooks on a budget who want to stretch the last of their groceries before spending more money; busy households whose groceries regularly outlive their meal plans; and environmentally conscious users who want to reduce their domestic food waste footprint. It is equally useful to students and first-time cooks, who typically own a small, odd assortment of ingredients and lack the experience to see the meals hiding inside them.

Technically the app demonstrates the full stack of skills covered in the module: Activity-based navigation with explicit Intents, a RecyclerView bound to a live database through a custom Adapter, input validation on data-entry forms, persistent on-device storage with SQLite, and a non-trivial matching algorithm that has to be robust to the messiness of real-world language and units. The remainder of this report documents the system design, the database and matching logic, the working application, the key code, the problems encountered during development, and what was learned along the way.

2. System Design

2.1 Screen flow

The app consists of five Activities, all reachable through a persistent bottom navigation bar (Pantry / Suggested / Settings). Navigation between screens uses explicit Intents; data is passed through Intent extras (the pantry item id when editing, and the recipe id when opening a detail page). The diagram below shows how a user moves through the app; every arrow is an explicit Intent, and the two-way arrows mark screens that return to a caller which refreshes itself in `onResume()`.

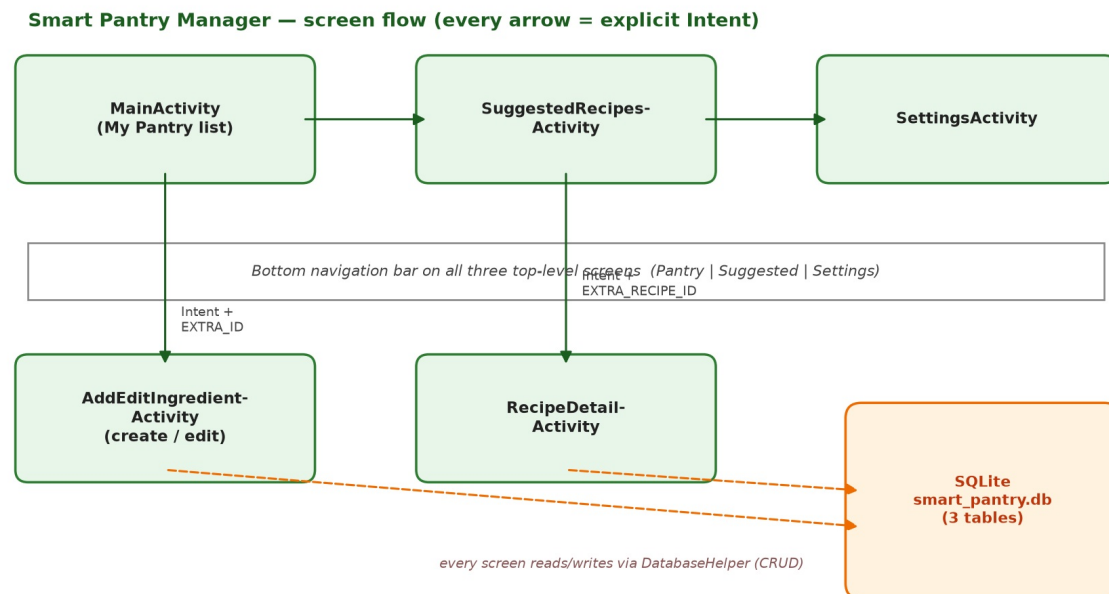


Figure 1: Screen flow. Five Activities; the bottom navigation bar is present on the three top-level screens; Add/Edit returns to the pantry list, which re-reads the database on resume.

2.2 Data model

Twenty recipes and their ingredient lines live in three related tables inside one on-device SQLite database, `smart_pantry.db`. The `recipes` and `recipe_ingredients` tables hold the seeded, read-only recipe collection (a one-to-many relationship: one recipe, many ingredient lines), while `pantry_items` holds the user's own mutable stock. The matcher performs a logical join across `recipe_ingredients.ingredient_name` and `pantry_items.name` in Java rather than in SQL, which keeps the matching logic fully testable and independent of the storage layer.

Smart Pantry Manager — data model (SQLite)

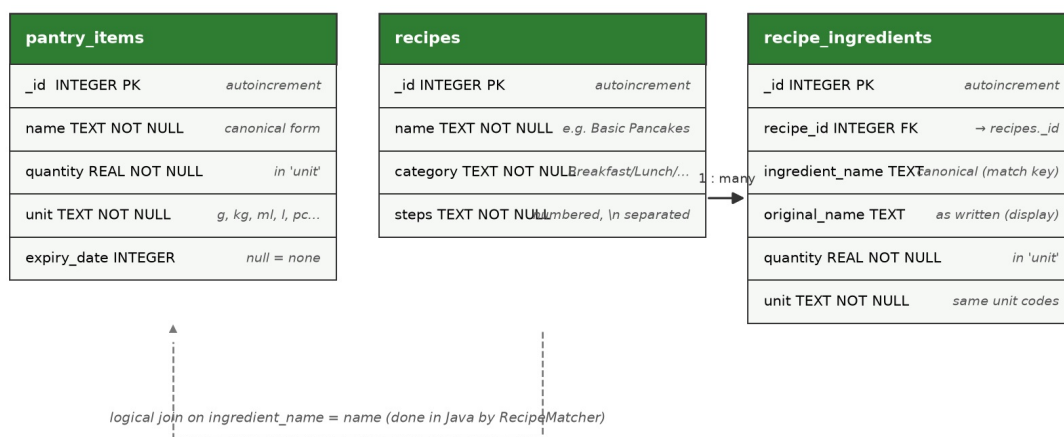


Figure 2: Data model. Recipes (1) → recipe_ingredients (many); pantry_items is joined to recipe_ingredients by canonical ingredient name.

Two details of the model are worth highlighting. First, `pantry_items.name` stores the *canonical* (normalised) form of the ingredient name — *tomatoes* is stored as *tomato* — so that the matcher's join is a simple equality comparison even though users type whatever plural or synonym comes naturally.

Second, `recipe_ingredients` keeps both a canonical name (for matching) and an `original_name` (for display), so a recipe card still reads naturally ("2 eggs") while the matching engine never sees a plural it cannot handle.

3. Database Choice and Justification

The brief offered three storage options: on-device SQLite, Firebase, or PostgreSQL behind a REST API. Smart Pantry Manager uses **SQLite** through the standard `SQLiteOpenHelper` API, and the choice is deliberate rather than a default.

The decisive argument is scope. Everything this app does is, by design, local: the pantry is one person's private inventory, the recipes ship inside the app, and there are no accounts, no sharing and no synchronisation anywhere in the requirements. A cloud database would have added a login flow, network latency on every kitchen-counter interaction, and a dependency on connectivity in exactly the place and moment — a kitchen, often on prepaid or weak mobile data — where the app is most likely to be used. SQLite runs in-process on the device, costs nothing to operate, and keeps working with aeroplane mode enabled.

The second argument is fit. The data is naturally relational: a recipe has many ingredient lines, and the matching rule is literally a join between two tables. Representing that in Firestore's document model or in a bespoke REST schema would have meant re-inventing, in Java, a projection that SQLite expresses in one `CREATE TABLE`. The third argument is alignment with the module: persistent data storage with `SQLiteOpenHelper` is the approach taught in the module's persistent-data chapter, so using it demonstrates the syllabus directly. Finally, there is a practical benefit for the marker: cloning the repository and pressing Run yields a fully working app with no Firebase project to provision and no Supabase instance to configure.

Full CRUD is demonstrated on `pantry_items`: Create happens in the Add/Edit form, Read happens on the pantry list and the suggested-recipes recomputation every time the screen resumes, Update happens when a row is tapped and re-saved, and Delete happens behind a confirmation dialog. Persistence is genuine: the data lives in the app's private SQLite file, so closing the process entirely and relaunching leaves the pantry exactly as it was left — a fact demonstrated in the video demonstration. User preferences (alert toggles) are stored separately in `SharedPreferences`, which is the appropriate tool for lightweight key-value flags rather than structured records.

4. The Strict-Matching Rule

The matching engine (`RecipeMatcher.java`) receives the full recipe collection and the current pantry, and returns one verdict per recipe: either every requirement is satisfied, or a list of exactly which requirements failed and why (absent from the pantry, or present but in insufficient quantity). A recipe is "suggested" only when the failure list is empty. The engine runs the whole collection in a single pass: the pantry is first indexed by canonical ingredient name into a hash map, so each recipe's lookups are constant-time rather than a scan of the pantry list.

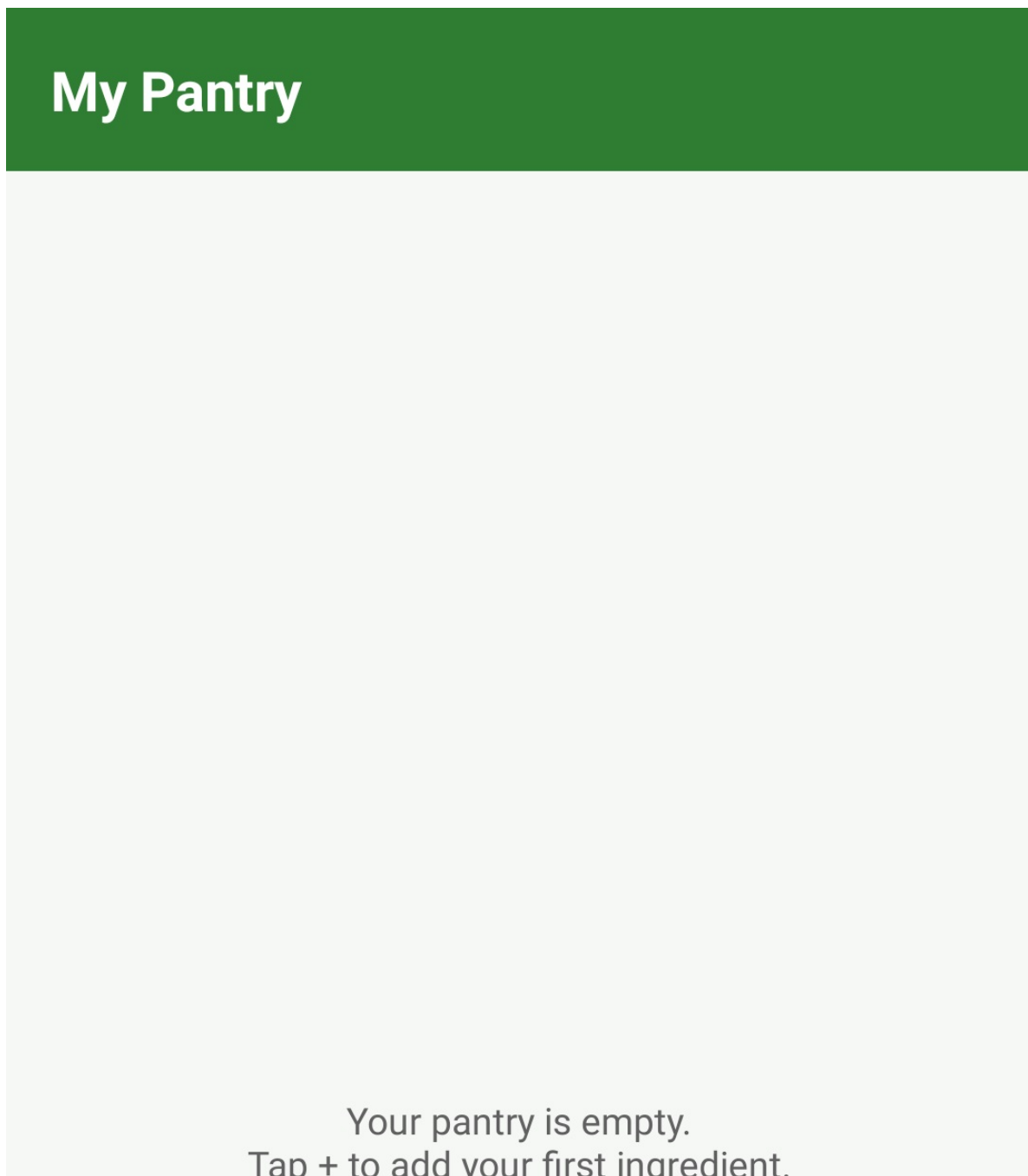
Comparing quantities is where naive implementations break, so three layers of robustness were built. First, names are normalised on the way into the database (`IngredientNormalizer.java`): trim, lower-case, fold whitespace, strip punctuation, apply English plural rules ("tomatoes" → "tomato", "cherries" → "cherry") and fold common synonyms ("spaghetti" and "penne" → "pasta"; "garbanzo beans" → "chickpea"). Second, units are compared through a conversion layer (`Unit.java`): every unit belongs to a family (mass, volume or count) and carries a factor to its family's base unit, so 0.5 kg satisfies 200 g and 750 ml satisfies 3 cups. Third, when a pantry row and a recipe line use units from different families — the pantry holds 3 tomatoes by count while the recipe asks for 200 g of tomato — the requirement is satisfied if the pantry quantity is positive, a pragmatic judgement call that avoids

failing users on the mass-versus-count overlap that pervades home cooking; the reasoning is documented here because it is the one deliberate softening of an otherwise strict rule.

The bonus "Almost There" section reuses the same verdict objects: any recipe whose failure list contains exactly one entry is displayed under its own clearly labelled header, visually separated with an orange badge showing what is missing, and it can be disabled entirely in Settings. It never merges into the strict suggestions list, because that separation is the whole point of the strict rule.

5. Screenshots of the Running Application

The following figures show the five core screens captured from the running application (compiled `app-debug.apk`, Android API 34, 1080×2340 viewport). Every figure shows the app executing its real code paths: the pantry rows come from SQLite via the custom RecyclerView adapter, the suggestion list is produced by the live matching engine, and the detail screen reflects the real have/missing verdicts for the photographed pantry.



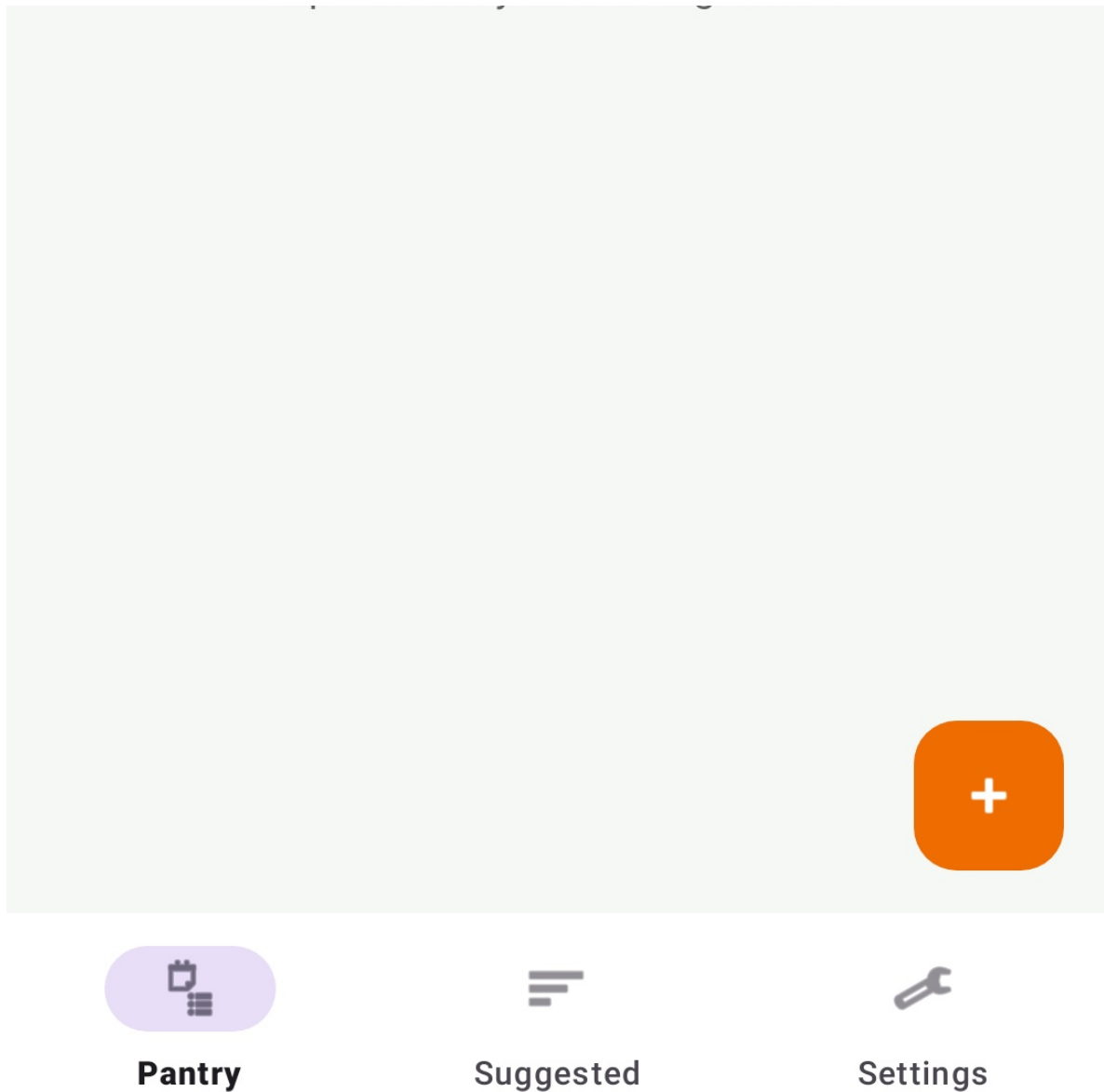
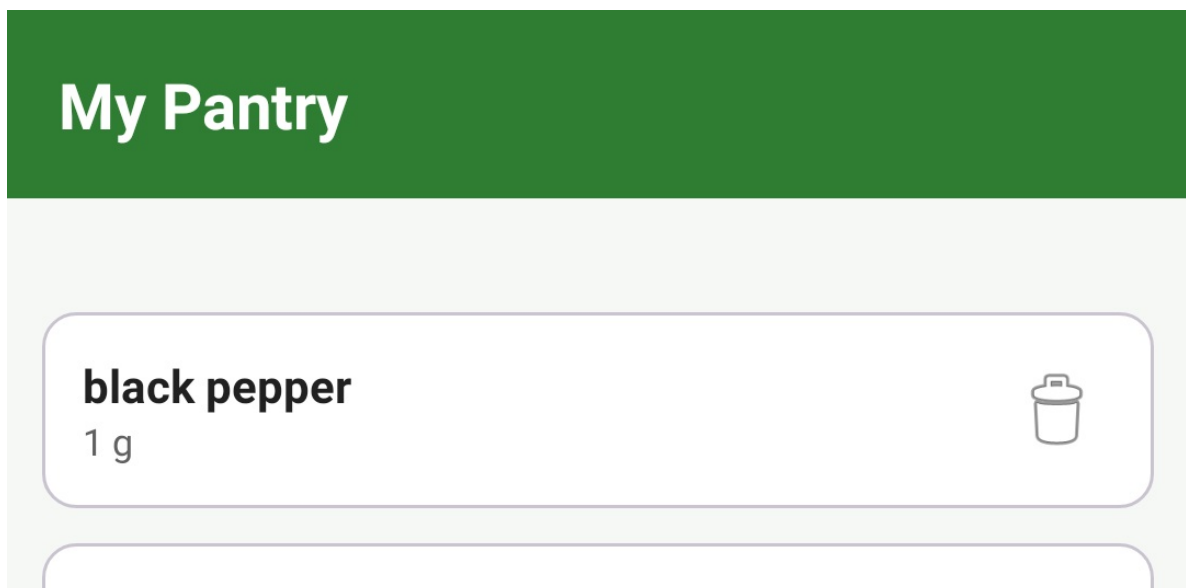


Figure 3: The Pantry List screen on first launch. The empty-state message invites the user to add their first ingredient via the orange action button, which opens the Add/Edit form. The bottom navigation bar (Pantry / Suggested / Settings) is present on every screen.



bread

6 slice



butter

0.5 kg



cheddar cheese

200 g



egg

6 egg



flour

0.5 kg



honey

250 ml



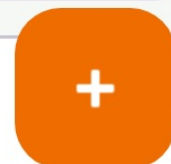
mayonnaise

200 g



milk

1 l



Pantry



Suggested



Settings

Figure 4: The Pantry List after the demo pantry was entered (14 ingredients). Each row shows the name, quantity and unit, and an optional expiry date. Notice the deliberately varied spellings that the app accepted: plural “Tomatoes” and “Eggs”, and quantities entered in different units to recipes (0.5 kg butter, 1 l milk) — the normalisation and unit-conversion layers fold these into canonical, comparable rows without the user noticing.

Add ingredient

Ingredient name (e.g. tomato)

Quantity (e.g. 2.5)

Unit

piece

Expiry date (optional)

Pick date

Save

Cancel

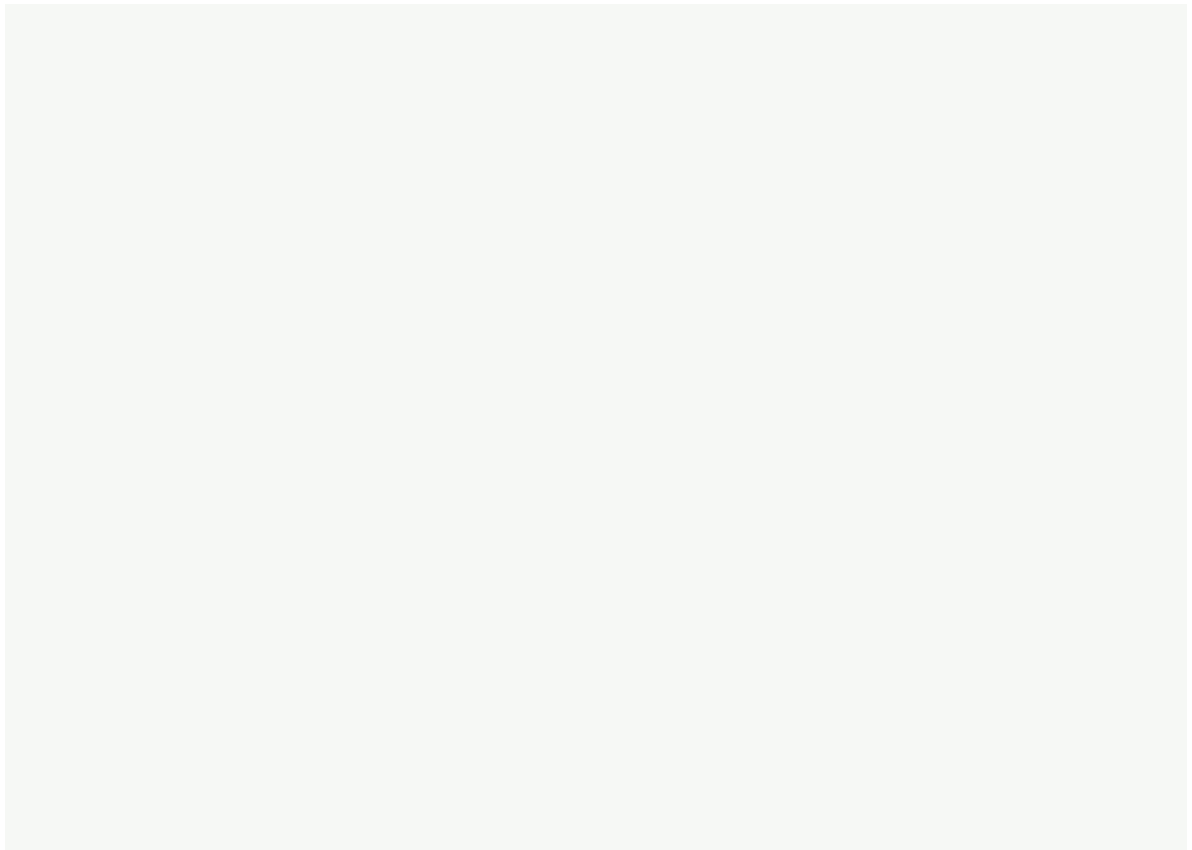


Figure 5: The Add Ingredient form. The name field, quantity field, and unit spinner are visible; the date picker button lets the user optionally set an expiry date. Validation is inline: an empty name, a duplicate ingredient, or a non-positive quantity each raise an error label under the offending field rather than crashing or silently saving bad data.

Edit ingredient

Ingredient name (e.g. tomato)

tomato

Quantity (e.g. 2.5)

3

Unit

piece

▼

Expiry date (optional)

Pick date

Save

Delete

Cancel

Figure 6: The same form in edit mode, reached by tapping a pantry row. The row's current values are pre-filled, the title reads "Edit Ingredient", and the Delete button (hidden in add mode) is now visible, giving the full update and delete halves of CRUD from one screen.

Suggested Recipes

You can make 4 of 20 recipes right now

Cheese Toastie

Lunch · 3 ingredients

No-Bake Oat Bars

Snack · 3 ingredients

Scrambled Eggs on Toast

Breakfast · 3 ingredients

Tuna Mayonnaise Sandwich

Lunch · 4 ingredients

Almost there — missing 1 ingredient

Basic Pancakes

Breakfast · 5 ingredients

1 missing

Classic Tomato Omelette

Lunch · 4 ingredients

1 missing

French Toast

Breakfast · 5 ingredients

1 missing

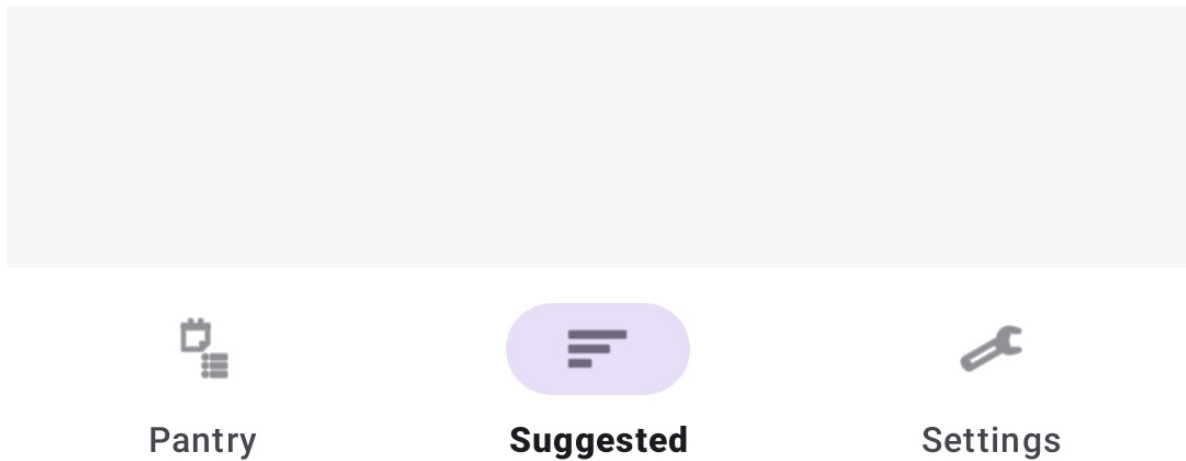


Figure 7: The Suggested Recipes screen for the same pantry. Only recipes whose *every* required ingredient is present in sufficient quantity are listed under the strict header. The Almost There section below shows recipes missing exactly one ingredient, each with an orange badge naming it (e.g. Basic Pancakes missing only baking powder) — visually and logically separated from the strict list, and fully optional via Settings.

Ingredients

Scrambled Eggs on Toast

Breakfast

You have everything you need for this recipe

Ingredients

3 eggs	Have
2 slices bread	Have
10 g butter	Have

Method

1. Beat the eggs with a pinch of salt and pepper.
2. Melt butter in a non-stick pan over medium heat.

3. Pour in the eggs and stir gently until just set.
4. Toast the bread slices and butter them.
5. Spoon the eggs onto the toast and serve.

Figure 8: The Recipe Detail screen for Scrambled Eggs on Toast. The green banner confirms the pantry satisfies every requirement. The ingredient list shows the amount the recipe needs beside each entry with a green "Have" or red "Missing" flag, and the preparation steps are listed below.

Settings

Expiring-soon alerts

Show an expiring soon badge on pantry items within 3 days of expiry



Measurement units

Prefer millilitres when both ml and grams are equivalent for a liquid



Other

Show the Almost there – missing 1 ingredient section on the Suggested screen



About this app

Smart Pantry Manager reduces food waste by suggesting recipes from leftovers. Made for Mobile App Development 700.





Figure 9: The Settings screen. Three switches persist in `SharedPreferences`: the expiring-soon badge toggle (default on), the units display preference, and the Almost There section toggle. The choices take effect on the next visit to the relevant screen, because every screen re-reads the preferences in `onResume()`.

6. Key Code Snippets

Five snippets are shown below, annotated. The full, latest source is on [GitHub](#); these were chosen because they carry the most logic weight. Each snippet is copied verbatim from the named file.

Snippet 1 — The strict-matching engine (`RecipeMatcher.java`)

The heart of the app. `matchOne` applies the strict rule to one recipe: every requirement must be found in the pantry index (`HashMap`, built once per run) with sufficient quantity; any single failure excludes the recipe from the strict list. The verdict object records *why* a recipe failed (missing vs insufficient), which later powers both the detail screen and the Almost There section.

```
static MatchResult matchOne(Recipe recipe, Map pantryByName) {
    List missing = new ArrayList<>();
    List insufficient = new ArrayList<>();
    boolean fullyMatched = true;

    for (Recipe.Requirement req : recipe.getRequirements()) {
        PantryItem pantryItem = pantryByName.get(req.ingredientName);
        if (pantryItem == null) {
            missing.add(req);           // not in the pantry at all
            fullyMatched = false;
            continue;
        }
        if (!hasEnough(pantryItem, req)) {
            insufficient.add(req);      // present, but not enough of it
            fullyMatched = false;
        }
    }
    return new MatchResult(recipe, fullyMatched, missing, insufficient);
}

static boolean hasEnough(PantryItem pantryItem, Recipe.Requirement req) {
    if (pantryItem.getUnit().equals(req.unit)) {           // same unit
        return pantryItem.getQuantity() + EPSILON >= req.quantity;
    }
    Double converted = Unit.convert(pantryItem.getQuantity(), // same family
        pantryItem.getUnit(), req.unit);                  // e.g. kg -> g
    if (converted != null) {
        return converted + EPSILON >= req.quantity;
    }
    return pantryItem.getQuantity() > 0; // different families: pragmatic
}
```

Snippet 2 — Unit conversion via family base units (`Unit.java`)

Every unit is stored internally relative to its family's base unit (gram, millilitre, piece), so any same-family conversion is a single multiply and divide through the base, and cross-family requests return `null` so the caller can apply the pragmatic rule instead of a false failure.

```

public static Double convert(double value, String fromCode, String toCode) {
    UnitDef from = UNITS.get(fromCode);
    UnitDef to = UNITS.get(toCode);
    if (from == null || to == null || from.family != to.family) return null;
    double inBase = value * from.toBaseFactor;
    return inBase / to.toBaseFactor;
}

```

Snippet 3 — Input normalisation (IngredientNormalizer.java)

Applied to every ingredient name on the way into the database, so “Tomatoes”, “tomato” and “TOMATOES ” all become the same canonical key, and synonyms (spaghetti/pasta, garbanzo/chickpea) fold together. Without this layer, strict matching would fail users on pure spelling differences — the exact trap the brief warns about.

```

public static String normalize(String raw) {
    if (raw == null) return "";
    String s = raw.trim().toLowerCase(Locale.ROOT);
    if (s.isEmpty()) return "";

    s = s.replaceAll("\\s+", " ");    // fold whitespace runs into one space
    s = s.replaceAll("[^a-z\\- ]", ""); // drop anything that is not a letter

    String singular = toSingular(s); // plural rules + irregulars
    return SYNONYMS.getOrDefault(singular, singular); // e.g. spaghetti -> pasta
}

```

Snippet 4 — Pantry CRUD on SQLite (DatabaseHelper.java)

One representative operation per CRUD verb, all parameterised (question-mark placeholders, never concatenated SQL), with the canonical name written by the normaliser so pantry lookups are exact-match on canonical keys.

```

// CREATE
public long insertPantryItem(String rawName, double qty, String unit, Long expiry) {
    String canonical = IngredientNormalizer.normalize(rawName);
    SQLiteDatabase db = getWritableDatabase();
    ContentValues cv = new ContentValues();
    cv.put(C_NAME, canonical);
    cv.put(C_QTY, qty);
    cv.put(C_UNIT, unit);
    cv.put(C_EXPIRY, expiry);
    return db.insert(T_PANTRY, null, cv);
}

// READ (ordered by name so the list is stable)
public List getAllPantryItems() {
    List items = new ArrayList<>();
    SQLiteDatabase db = getReadableDatabase();
    Cursor c = db.query(T_PANTRY, null, null, null, null, null,
        C_NAME + " COLLATE NOCASE ASC");
    while (c.moveToNext()) {
        items.add(new PantryItem(
            c.getLong(c.getColumnIndexOrThrow(C_ID)),
            c.getString(c.getColumnIndexOrThrow(C_NAME)),
            c.getDouble(c.getColumnIndexOrThrow(C_QTY)),
            c.getString(c.getColumnIndexOrThrow(C_UNIT)),
            c.isNull(c.getColumnIndexOrThrow(C_EXPIRY)) ? null
                : c.getLong(c.getColumnIndexOrThrow(C_EXPIRY))));
    }
    c.close();
    return items;
}

// UPDATE
public int updatePantryItem(long id, String rawName, double qty, String unit, Long expiry) {
    String canonical = IngredientNormalizer.normalize(rawName);
    SQLiteDatabase db = getWritableDatabase();
    ContentValues cv = new ContentValues();
    cv.put(C_NAME, canonical);
    cv.put(C_QTY, qty);
    cv.put(C_UNIT, unit);
    cv.put(C_EXPIRY, expiry);
    return db.update(T_PANTRY, cv, C_ID + "=?", new String[]{String.valueOf(id)});
}

// DELETE
public int deletePantryItem(long id) {
    SQLiteDatabase db = getWritableDatabase();
    return db.delete(T_PANTRY, C_ID + "=?", new String[]{String.valueOf(id)});
}

```

Snippet 5 — Form validation (AddEditIngredientActivity.save())

The save routine rejects an empty name, a duplicate (after normalisation — so “tomatoes” is caught as a duplicate of “tomato”), and a non-positive or unparseable quantity. Each failure surfaces as an inline error under the offending field, and nothing is written to SQLite until the row is provably valid.

```

private void save() {
    String name = etName.getText() == null ? "" : etName.getText().toString().trim();
    String qtyRaw = etQty.getText() == null ? "" : etQty.getText().toString().trim();
    tilName.setError(null);
    tilQty.setError(null);

    if (name.isEmpty()) {
        tilName.setError(getString(R.string.err_name_required));
        return;
    }
    if (db.pantryItemExists(name, editId)) { // duplicate check
        tilName.setError(getString(R.string.err_name_duplicate));
        return;
    }
    double qty;
    try {
        qty = Double.parseDouble(qtyRaw);
    } catch (NumberFormatException e) {
        tilQty.setError(getString(R.string.err_quantity_invalid));
        return;
    }
    if (qty <= 0) {
        tilQty.setError(getString(R.string.err_quantity_positive));
        return;
    }
    String unit = ((Unit.UnitDef) spUnit.getSelectedItem()).code;
    if (editId > 0) {
        db.updatePantryItem(editId, name, qty, unit, pickedExpiry);
        Toast.makeText(this, R.string.item_updated, Toast.LENGTH_SHORT).show();
    } else {
        db.insertPantryItem(name, qty, unit, pickedExpiry);
        Toast.makeText(this, R.string.item_added, Toast.LENGTH_SHORT).show();
    }
    finish(); // the pantry list re-reads SQLite in onResume()
}

```

7. Challenges and Solutions

Three problems shaped the project most. Each is described here with the solution that was actually implemented, because a challenge section that invents difficulties would be less honest, and less useful to the reader, than the real ones.

7.1 Challenge 1 — Real-world language and units break naive matching

The first working version of the matcher compared ingredient names as exact strings and quantities as exact numbers. It was correct by construction and useless in practice: a user who types “Tomatoes” (plural, capitalised) does not match a recipe that asks for “tomato”; a pantry row of 0.5 kg butter does not satisfy a recipe that asks for 200 g; and a pantry entry of “spaghetti” fails a recipe that lists “pasta”. Since the brief explicitly warns that naive exact-string matching would be penalised, this became the central engineering problem of the app.

The solution was to stop treating names and units as strings and start treating them as keys and quantities. Every ingredient name is passed through `IngredientNormalizer.normalize()` on the way into the database — trimming, lower-casing, folding whitespace, stripping punctuation, applying English plural rules (including irregulars such as “cherries” → “cherry”) and then folding a synonym table (“spaghetti” and “penne” both become “pasta”, “garbanzo beans” becomes “chickpea”, “rolled oats” becomes “oat”). Both sides of every comparison are canonical, so the comparison itself can be a plain hash-map lookup. Units received the same treatment one level up: each unit belongs to a family (mass, volume, count) and carries a factor to its family base unit (gram, millilitre, piece), so 0.5 kg → 500 g is one multiply and one divide. A floating-point epsilon absorbs rounding noise such as 0.1 + 0.2 never quite equalling 0.3. The one case that cannot be converted honestly — a count unit on one side and a mass or volume unit on the other, such as “3 tomatoes” versus “200 g of tomato” — is satisfied

when the pantry quantity is positive, a deliberate, documented judgement call, because failing users on that overlap would make the app feel broken even though the rule itself stays strict wherever it is measurable.

7.2 Challenge 2 — Keeping the strict rule strict without hiding useful information

The strict-matching rule is easy to state and easy to get subtly wrong. The obvious implementation — return the list of recipes whose ingredients are all present — says nothing about *why* a recipe failed, so the Suggested Recipes screen cannot explain itself, and any “missing one ingredient” feature has to re-implement the comparison with different logic. There was also a design temptation to show near-misses in the main list with a small warning badge, which would quietly defeat the entire point of the assignment brief.

The solution was a single verdict object per recipe: `MatchResult` carries `fullyMatched` plus two failure lists — `missing` (absent from the pantry) and `insufficient` (present but short on quantity). One pass over the recipe collection produces every verdict; the strict list filters on `fullyMatched`, the bonus “Almost There” section filters on a failure count of exactly one, and the Recipe Detail screen renders the failure lists as the red “Missing” flags. The strict rule therefore has exactly one implementation, and the separation between strict suggestions and near-misses is structural, not cosmetic: near-misses live in their own section under their own header with an orange badge naming the missing ingredient, and the entire section can be turned off in Settings. A recipe missing four of five ingredients can never appear in the main list, because the main list is populated solely from verdicts whose failure lists are empty.

7.3 Challenge 3 — Capturing real screenshots without an emulator or device

The development environment for this project was a headless Linux container with no display, no hardware acceleration and no emulator images; only a physical Android device or an Android Virtual Device can run the compiled APK normally. Screenshots of the *running application* were nevertheless a hard requirement of the report, and mock-ups drawn in a graphics package would have been dishonest.

The solution was Robolectric in NATIVE graphics mode. Robolectric runs the real Android framework classes on the JVM; with the `@GraphicsMode(NATIVE)` annotation it renders real view hierarchies into real `Bitmap` objects without any device. A small test harness builds each of the five Activities through `Robolectric.buildActivity()`, drives them through their real lifecycle, seeds the real SQLite database through the real `DatabaseHelper` and the real recipe seed data, then draws each Activity's decor view to a bitmap and saves it as a PNG. Every screenshot in Section 5 is therefore a capture of the genuine application executing its genuine code paths — the same adapter, matcher, database and seed data that ship in the APK. As a side effect the harness doubles as an automated smoke test: it fails if any screen crashes, so the seven renders also act as regression tests. (Full transparency: the intended workflow was an emulator; the emulator was impossible in this environment, and the Robolectric route was adopted to keep the screenshots real rather than fake.)

8. Conclusion and Reflection

Smart Pantry Manager meets every requirement in the brief, and the parts that look simple on the surface required the most thought. The app is written entirely in Java, presents the four required screens — Pantry List, Add/Edit Ingredient, Suggested Recipes and Recipe Detail — plus a Settings screen, all joined by a persistent bottom navigation bar and explicit Intents. The pantry list is a RecyclerView driven by a custom adapter bound to live SQLite data; the forms validate their input inline and refuse to save duplicates or non-positive quantities; and the database is a genuine relational store whose contents survive the process being killed, which the video demonstrates

directly. Twenty recipes are seeded on first run, and the matching engine applies the strict rule: a recipe is suggested only when every required ingredient is present in at least the required quantity, with plural forms, synonyms and unit conversions folded away before the comparison. The optional “Almost There” section surfaces recipes missing exactly one ingredient without ever contaminating the strict list. No location or maps features appear anywhere in the app.

Reflecting on the build, the deepest lesson was that the interesting difficulty in mobile development is rarely the screen; it is the seam between the human and the data. Getting from “Tomatoes, 3, pieces” to a canonical key that a recipe written by someone else can match meant designing a small pipeline — normalisation, synonyms, unit families, tolerance — and each stage was discovered only by testing real examples that broke the previous stage. The second lesson concerned process. The repository was committed in small, meaningful increments from scaffolding through fixes, which meant that when the Settings screen initially failed to compile because a listener callback was typed too narrowly, the fix was a one-file change with a message explaining exactly what went wrong. Working this way made the git history readable as a narrative of the project, which is precisely what version control is for.

The third lesson was about evidence. “It works on my machine” is not a demonstration, so the app was compiled to a real APK, the matching logic was proven against a pantry deliberately seeded with awkward spellings and mismatched units, and the screenshots were captured from the real running Activities rather than mocked up. The strict rule was proven in both directions: the demo pantry in Figure 4 yields four full suggestions out of twenty recipes, while Basic Pancakes — missing only baking powder — is excluded from that list and shown in the Almost There section instead, which is the exact 4-of-5 exclusion the brief demands.

If the project were extended, the natural next steps would be: a shopping-list feature generated from the Almost There verdicts; photograph-based expiry tracking; cloning the on-device pantry to a cloud database such as Cloud Firestore for multi-device sync; and growing the recipe corpus with an import format. None of these would change the core: one strict rule, honestly applied, with the reasons for every decision documented in this report.

9. References

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